Elie Chadaida

Centre for Macroeconomics, London School of Economics
elie.chadaida.n@gmail.com | +33 6 77 52 76 91 | Website | LinkedIn

Research Interests

Macroeconomics and behavioural economics: heterogeneity in households' beliefs and expectations formation, housing, distributional effects of monetary policy; computational methods for heterogeneous-agent models.

Education

London School of Economics and Political Science, London, United Kingdom 08/2023 – 06/2024
MSc in Finance and Economics, *Distinction*

- Courses: Macroeconomics – Microeconomics – Financial Econometrics – Stochastic Calculus – Financial Economics – Financial Engineering – Fixed Income Securities and Credit Markets

CentraleSupélec, Paris, France 09/2020 – 06/2024
MSc and BSc (*Diplôme d'Ingénieur*), Mathematical and Financial Modelling, *First-Class Honours*

- Second-ranked engineering school in France (*Grandes Écoles* track); double degree with LSE
- Courses: Fundamental and Applied Mathematics and Statistics – Optimisation – Computer Science, Algorithmics, and Machine Learning – Economics and Corporate Finance – Physics and Chemistry

Lycée Sainte-Geneviève, Versailles, France 08/2018 – 07/2020
Classes préparatoires (PCSI/PC*), *First-Class Honours*

- Two-year intensive programme in mathematics, physics and chemistry preparing for the national competitive entrance exams to the French engineering *Grandes Écoles*

Summer School: Tools for Macroeconomists – The Essentials and Advanced 08/2026
Profs. Petr Sedláček and Pontus Rendahl – perturbation methods and Dynare, Kalman filter and Bayesian estimation of DSGE models, global solution methods, heterogeneous-agent models

Research Positions

Pre-Doctoral Research Assistant, London School of Economics, London, United Kingdom 09/2025 – Now
Centre for Macroeconomics – Profs. Matthias Doepke & Ethan Ilzetzki

Distributional effects of monetary policy (Doepke)

- Ported a heterogeneous-agent OLG model of inflation redistribution (housing tenure choice, nominal assets) from MATLAB to modular Python: DC-EGM with logit taste shocks, steady-state calibration, and transition dynamics after an inflation shock with market-clearing house prices
- Extending the Doepke–Schneider (2006) sectoral net nominal positions to the latest Federal Reserve Financial Accounts (Z.1) releases: remapped re-coded series and rebuilt the MATLAB pipeline as a metadata-driven Python pipeline

Government-directed R&D and Innovation: Evidence from NASA Budget Requests (Handlan, Ilzetzki & Juhász)

- Built a DuckDB database linking US patents (1945–2000) to technology classes (CPC/USPC), patent and paper citations, firm values and government-interest flags, and the class × year regression panel
- Designed an LLM pipeline that reads NASA congressional budget requests (1961–1980), extracts their technical problems and ranks the relevant patent classes to allocate programme budgets; benchmarked it against BERT-embedding and hybrid methods, with prompt and sensitivity analyses
- Estimated staggered difference-in-differences (Callaway–Sant'Anna, IPW and doubly robust, with propensity-score trimming), continuous-treatment DiD (de Chaisemartin–D'Haultfœuille), local-projection DiD and knowledge production functions in Stata and R; parallelised the estimations for a SLURM computing cluster
- Produced chartbooks, automated LaTeX write-ups and presentation figures, including for the SED 2026

Other projects

- *Gender and fertility (Doepke):* assembled cross-country data on childlessness, marriage and gender attitudes (IPUMS, UN World Marriage Data, World Values Survey) for Doepke & Kindermann (2026), "Gender Relations and Fertility: An Intricate Relationship" (acknowledged), and a project on men's stake in women's empowerment
- *Wages, work and war bonds in WWII (Ilzetzki):* reviewed and documented archival labour-market and bond data sources and digitised state-level war-bond sales
- *Policy (Ilzetzki):* prepared a literature review and comparative-evidence briefing on fiscal rules and fiscal councils for the House of Lords Economic Affairs Committee; ran cross-country regressions of house price-to-rent ratios
- *Centre for Macroeconomics:* wrote model solutions for the pre-doctoral recruitment task and assessed applicants

Other Positions

European Central Bank, Frankfurt am Main, Germany 09/2024 – 08/2025
Trainee, Market Operations Analysis Division

- Managed the databases used to monitor monetary policy implementation; analysed the likely effects of changes to the minimum reserve requirements framework and presented the results to different divisions
- Network analysis of financial interconnection between euro-area banks (Python, SQL); research project on the AI exposure of EU employment (see *Other Research Projects*); taught three SQL and Python classes to economist colleagues

Roland Berger, Paris & Amsterdam 09/2022 – 12/2022 & 05/2023 – 07/2023
Strategy Analyst Intern

- Built and presented financial and economic models comparing technologies and strategies; research on regulation and competition in financial services and European industrials

HomeExchange, Paris 01/2023 – 04/2023
Data Analyst Intern

- Modelled part of the company's indirect carbon emissions from 15,000+ survey responses and automated the calculation in VBA

Research Projects

Direction over Size: Sign Dominance in House-Price Expectations Draft, 11/2026
Single-authored – draft available on my website

- Uses Michigan Surveys of Consumers and New York Fed SCE Housing Survey microdata to show that households form short-term house-price expectations mainly in response to the sign of perceived past price growth rather than its size: expectations jump when the perceived direction of prices turns
- Estimates an interval ("band") model of the answers by maximum likelihood with wild cluster bootstrap inference; a partial-equilibrium lifecycle model with a rent-or-buy decision, in which households hold the measured beliefs, shows that purchases are delayed and bunch when prices turn

Other Research Projects

AI and Productivity Gains in the European Union Public and Private Sectors 2025
with Emilien Alessandrini and Maximilian Freier (European Central Bank)

- Combines a task-based measure of occupational exposure to generative AI with EU-LFS and EU-SILC microdata to measure the AI exposure of public and private employment across EU countries and the potential fiscal savings; built the data pipeline and contributed to the analytical framework and the interpretation of results

GARCH and LMMR Forecasts of Implied Volatility: Economic Synergies 2024
MSc thesis, London School of Economics (Financial Engineering)

- Developed a daily trading environment with portfolio updates and PnL calculations, using GARCH and LMMR forecasts to implement signal trading strategies on S&P 500 options

Additional Information

Programming: Python (heterogeneous-agent models, data pipelines, econometrics, machine learning) – Stata (panel data, difference-in-differences) – R (econometrics) – MATLAB and Dynare (macroeconomic models) – SQL and DuckDB (large databases) – C# – VBA – JavaScript – HTML/CSS
Tools: LaTeX, Git, SLURM computing clusters, LLM APIs, Excel, PowerPoint, Power BI, Tableau
Test scores: GRE General (01/2026): Quantitative 170/170 (89th percentile), Verbal 153, Analytical Writing 4.0 – TOEFL iBT (09/2026): 118/120 (CEFR C2)
Teaching: Private tutoring in mathematics and physics (high-school and bachelor's level) and in finance (master's level), [YEARS]; SQL and Python classes for economists at the ECB
Languages: French (native) – English (fluent) – Arabic (B2) – Spanish (B1)
Interests: Swimming, running, music, cooking

References

Prof. Matthias Doepke
Nicholas Kaldor Professor of Economics
London School of Economics
m.doepke@lse.ac.uk
Website

Prof. Ethan Ilzetzki
Associate Professor of Economics
London School of Economics
e.ilzetzki@lse.ac.uk
Website

Dr Jean-Pierre Zigrand
Associate Professor of Finance
London School of Economics
j.p.zigrand@lse.ac.uk
Website